

## Chapter - 14



### KEY CONCEPT

Breathing is a mechanical process of taking in  $O_2$  and expelling  $CO_2$  whereas exchange of gases occurs at cellular level.

Life sustains where breath begins

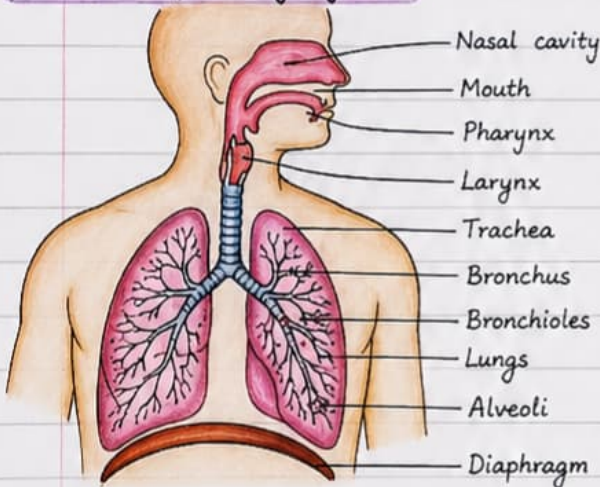
# Breathing and Exchange of Gases

All living organisms need **energy** to carry out life processes. This energy is released during the **oxidation** of organic food. The ultimate source of energy for most organisms is **oxygen**.

## 1. BREATHING

**Breathing** is a physical process which involves inhalation (taking in  $O_2$ -rich air) and exhalation (releasing  $CO_2$ -rich air). It does not involve the use or release of energy.

### Human Respiratory System



### Pathway of Air

Nasal cavity  
↓  
Pharynx  
↓  
Larynx  
↓  
Trachea  
↓  
Bronchi  
↓  
Bronchioles  
↓  
Alveoli

Air is filtered, warmed and moisturized in the nasal cavity.

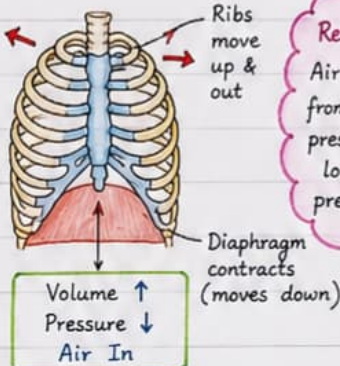
### Functions of Respiratory System

- It brings  $O_2$  in the body.
- It removes  $CO_2$  from the body.
- It helps in maintaining acid-base balance.
- It helps in phonation (voice production) and olfaction (smell).

## 2. MECHANISM OF BREATHING

### (A) INHALATION

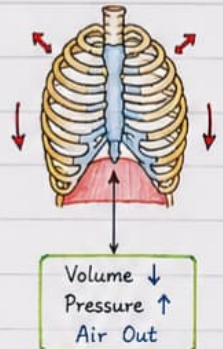
- Diaphragm contracts and moves down.
- Ribs move upward and outward.
- Thoracic cavity volume increases.
- Pressure inside lungs decreases.
- Air rushes in.



**Remember**  
Air moves from higher pressure to lower pressure.

### (B) EXHALATION

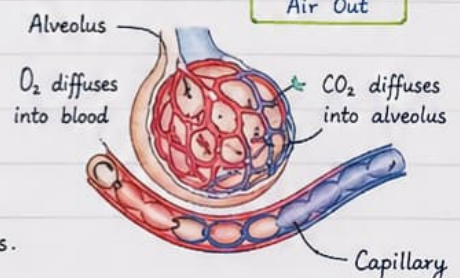
- Diaphragm relaxes and moves up.
- Ribs move downward and inward.
- Thoracic cavity volume decreases.
- Pressure inside lungs increases.
- Air is pushed out.



## 3. EXCHANGE OF GASES

Exchange of gases refers to the diffusion of  $O_2$  and  $CO_2$  between the respiratory surfaces (alveoli) and blood.

- External respiration**: Exchange of gases between alveoli and blood.
- Internal respiration**: Exchange of gases between blood and body cells.



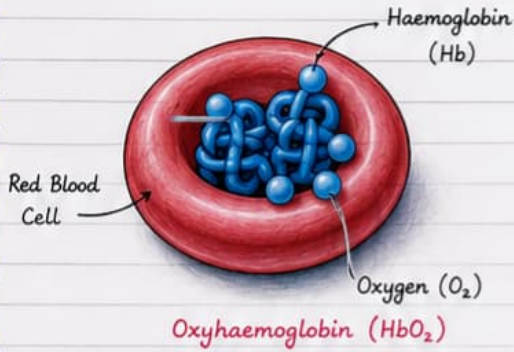


## 4. TRANSPORT OF GASES

### (A) TRANSPORT OF OXYGEN

- ★ After diffusion into blood,  $O_2$  is carried to tissues mainly by **RBCs**.
- ★ Binding of  $O_2$  with haemoglobin (Hb) forms **oxyhaemoglobin**.
- ★ Only a small amount of  $O_2$  ( $\approx 2\%$ ) dissolves in plasma.

#### OXYGEN TRANSPORT



Hb has very high affinity for  $O_2$  in lungs but low affinity in tissues. ♥

### (B) TRANSPORT OF CARBON DIOXIDE

$CO_2$  is transported in blood in three forms :

1. As **bicarbonate ions** ( $HCO_3^-$ ) - 70%
2. As **carbamino compounds** - 20-23%
3. Dissolved in plasma - 7-10%

#### HOW BICARBONATE IS FORMED ?

$CO_2$  diffuses into RBC  $\rightarrow$  combines with  $H_2O$  (with enzyme **carbonic anhydrase**)  $\rightarrow$  forms  $H_2CO_3 \rightarrow$  dissociates into  $H^+ + HCO_3^-$   
 $HCO_3^-$  moves out to plasma in exchange of  $Cl^-$  (chloride shift).



- As bicarbonate ions ( $HCO_3^-$ )
- As carbamino compounds
- Dissolved in plasma

★ In tissues :  $CO_2$  is picked up by blood.  
 In lungs :  $CO_2$  is released from blood.

## 5. DIFFERENCE BETWEEN EXTERNAL AND INTERNAL RESPIRATION

| Feature             | External Respiration                                                                                                                                          | Internal Respiration                                                                                                                                      |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Place               | Between alveoli (lungs) and blood.                                                                                                                            | Between blood and body cells (tissues).                                                                                                                   |
| Process             | <ul style="list-style-type: none"> <li><math>O_2</math> diffuses from alveoli to blood.</li> <li><math>CO_2</math> diffuses from blood to alveoli.</li> </ul> | <ul style="list-style-type: none"> <li><math>O_2</math> diffuses from blood to cells.</li> <li><math>CO_2</math> diffuses from cells to blood.</li> </ul> |
| Purpose             | To oxygenate blood.                                                                                                                                           | To supply $O_2$ to cells and remove $CO_2$ from cells.                                                                                                    |
| Surface involved    | Respiratory membrane (alveolar epithelium + capillary).                                                                                                       | Tissue fluid and cell membrane.                                                                                                                           |
| Direction of $O_2$  | Air $\rightarrow$ Alveoli $\rightarrow$ Blood                                                                                                                 | Blood $\rightarrow$ Cells                                                                                                                                 |
| Direction of $CO_2$ | Blood $\rightarrow$ Alveoli $\rightarrow$ Air                                                                                                                 | Cells $\rightarrow$ Blood                                                                                                                                 |

#### Key Point

External respiration is a part of pulmonary respiration. ♥

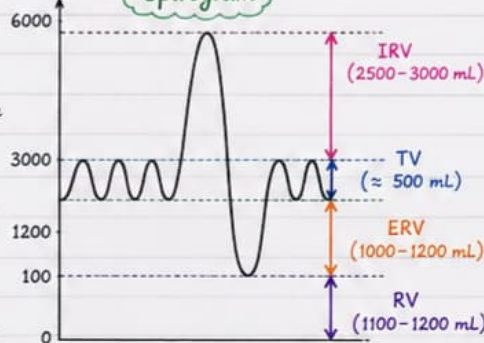
Internal respiration is also called tissue respiration or cellular respiration. ♥

## 5. RESPIRATORY VOLUMES AND CAPACITIES

- ★ **Tidal Volume (TV)** : Volume of air inhaled or exhaled during normal breathing. ( $\approx 500$  mL)
- ★ **Inspiratory Reserve Volume (IRV)** : Additional volume of air that can be inhaled with maximum effort. ( $\approx 2500-3000$  mL)
- ★ **Expiratory Reserve Volume (ERV)** : Additional volume of air that can be exhaled with maximum effort after a normal exhalation. ( $\approx 1000-1200$  mL)
- ★ **Residual Volume (RV)** : Volume of air remaining in the lungs even after maximum exhalation. ( $\approx 1100-1200$  mL)

★ Residual volume cannot be measured by spirometer. ♥

#### Spirogram



$$\begin{aligned}
 IC \text{ (Inspiratory Capacity)} &= TV + IRV \\
 &\approx 3000-3500 \text{ mL} \\
 FRC \text{ (Functional Residual Capacity)} &= ERV + RV \\
 &\approx 2100-2300 \text{ mL} \\
 VC \text{ (Vital Capacity)} &= IRV + TV + ERV \\
 &\approx 4500-5000 \text{ mL} \\
 TLC \text{ (Total Lung Capacity)} &= VC + RV \\
 &\approx 5600-6000 \text{ mL}
 \end{aligned}$$

#### Remember

Vital Capacity (VC) is the maximum volume of air that can be exhaled after maximum inhalation.

Total Lung Capacity (TLC) is the total volume of air in lungs after maximum inhalation. ♥

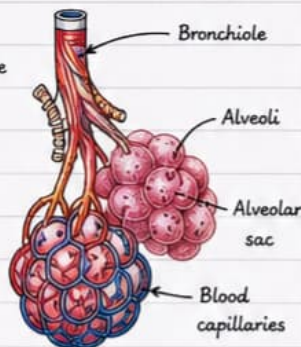


## 6. GASEOUS EXCHANGE IN HUMAN BEINGS

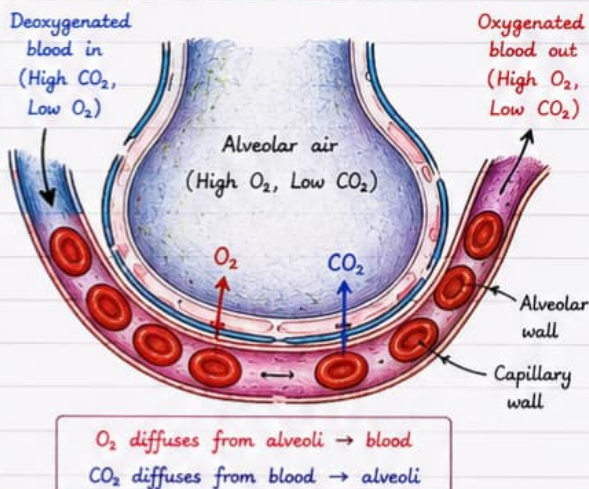
Gaseous exchange occurs in the alveoli of lungs through the process of **diffusion**.

### STRUCTURE OF ALVEOLUS

- Alveoli are tiny, thin-walled, balloon-like structures present in large numbers.
- They provide a very large surface area for exchange of gases.
- Walls are one cell thick (squamous epithelium) and covered with moist surfactant.
- Surrounded by a network of blood capillaries.



### GAS EXCHANGE AT ALVEOLI

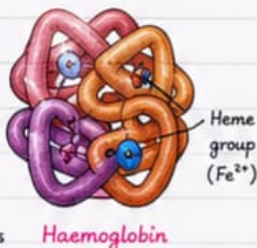


## 7. RESPIRATORY PIGMENTS

- Respiratory pigments are the molecules which bind and transport  $O_2$ .

### (A) HAEMOGLOBIN (Hb)

- Present in RBCs.
- Each Hb molecule has 4 polypeptide chains and each chain has one heme group with  $Fe^{2+}$  at the centre.
- One Hb molecule can carry 4  $O_2$  molecules.
- Forms oxyhaemoglobin in lungs and releases  $O_2$  in tissues.



### (B) HAEMOCYANIN

- Found in some invertebrates (e.g., Pila, Limulus).
- Contains copper ( $Cu^{2+}$ ) instead of iron.
- Turns blue when oxygenated.



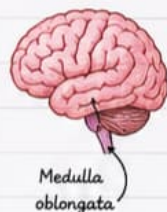
Affinity of Hb for  $O_2$  is high in lungs and low in tissues – helps in easy loading and unloading of  $O_2$ .

## 8. REGULATION OF BREATHING

- Breathing is an involuntary process.
- It is regulated by the respiratory centres in the **medulla oblongata**.

### RESPIRATORY CENTRE

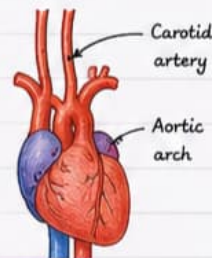
- Located in the medulla oblongata.
- It receives impulses (feedback) from chemoreceptors.
- It sends impulses to respiratory muscles (diaphragm and intercostals) to change the rate and depth of breathing.



### CHEMORECEPTORS

Sensitive to changes in  $CO_2$ ,  $O_2$  and pH of blood.

- Central chemoreceptors** – in medulla oblongata (respond to  $CO_2$ ).
- Peripheral chemoreceptors** – in carotid arteries and aortic arch (respond to  $CO_2$ ,  $O_2$  and pH).



## 9. FACTORS AFFECTING THE RATE OF BREATHING

| Factor                       | Effect                               | Reason                                          |
|------------------------------|--------------------------------------|-------------------------------------------------|
| Exercise / Physical activity | Increases ↑                          | More energy required → more $O_2$ needed        |
| Age                          | Decreases with increase in age ↓     | Metabolic rate decreases                        |
| Temperature                  | Increases with rise in temperature ↑ | More energy needed to maintain body temperature |
| Emotions (fear, anger)       | Increases ↑                          | Due to increased metabolic activity             |
| Altitude (high altitude)     | Increases ↑                          | Low $O_2$ availability in air                   |
| Diseases (fever, asthma)     | Increases ↑                          | Body needs more $O_2$                           |
| Sleeping                     | Decreases ↓                          | Less energy requirement                         |

### Key Points

- Breathing is a mechanical process.
- Exchange of gases occurs in alveoli by diffusion.
- $O_2$  is carried by haemoglobin in RBCs.
- $CO_2$  is transported mainly as bicarbonate ions.
- Breathing is regulated by respiratory centre in medulla oblongata.
- Chemoreceptors monitor  $CO_2$ ,  $O_2$  and pH to control breathing rate.

### SUMMARY FLOW

Air Inhaled (Nose) → Alveoli ( $O_2$  in,  $CO_2$  out) → Blood (Transport) → Tissues ( $O_2$  used,  $CO_2$  produced) →  $CO_2$  back to lungs (Exhaled)

Breathe in knowledge,  
Breathe out success!



## 10. RESPIRATORY DISORDERS

Smoking, pollution, infections and allergens can affect the respiratory system.

| Disorder           | Cause                                     | Symptoms / Effects                           | Prevention / Control                 |
|--------------------|-------------------------------------------|----------------------------------------------|--------------------------------------|
| <b>Asthma</b>      | Allergens, dust, smoke, pollen            | Breathlessness, wheezing, coughing           | Avoid allergens, medicines, inhalers |
| <b>Bronchitis</b>  | Infection, smoking, air pollution         | Inflammation of bronchi, cough, excess mucus | Avoid smoking, clean air, medication |
| <b>Emphysema</b>   | Smoking, long term exposure to pollutants | Damage to alveoli, breathlessness            | Quit smoking, oxygen therapy         |
| <b>Pneumonia</b>   | Bacterial/viral infection                 | Fever, chest pain, difficulty in breathing   | Vaccination, hygiene, medicines      |
| <b>Lung Cancer</b> | Smoking, chemicals, pollution             | Persistent cough, weight loss, chest pain    | Avoid smoking, early detection       |

## 11. ARTIFICIAL RESPIRATION

When breathing stops due to drowning, electric shock, or suffocation, artificial respiration can save life.

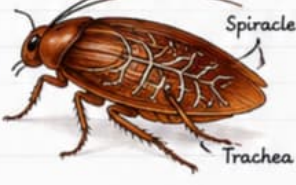
- ★ **Aim** : To supply  $O_2$  to lungs and remove  $CO_2$  until normal breathing returns.
- ★ **Methods** : Mouth to mouth, Mouth to nose, Holger Nielsen method, etc.
- ★ **Principle** : Repeated inflation and deflation of lungs.



Clean airway first, then give rescue breathes.

## 12. BREATHING IN OTHER ORGANISMS

Different organisms have different respiratory organs suited to their habitat and body structure.

| (A) AMOEBA                                                                        | (B) EARTHWORM                                                                      | (C) INSECTS (e.g., Cockroach)                                                      | (D) FISH                                                                             | (E) PLANTS                                                                           |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |  |   |  |
| Gas exchange occurs through the cell surface by diffusion.                        | Moist skin with rich blood supply acts as respiratory surface.                     | Have tracheal system. Air enters through spiracles and travels in tracheae.        | Extract $O_2$ dissolved in water through gills. Water flows over gills for exchange. | Exchange of gases through stomata, lenticels and root hairs by diffusion.            |

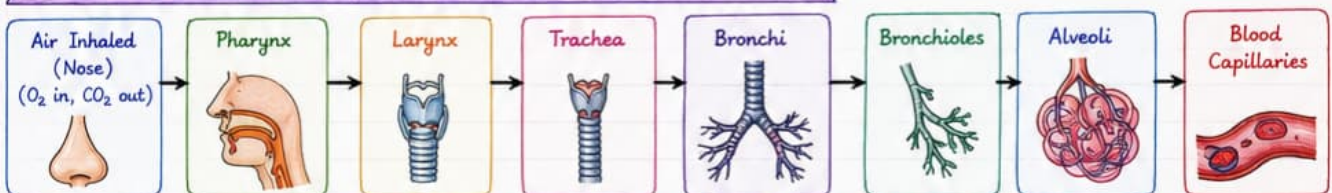
## 13. IMPORTANCE OF BREATHING AND EXCHANGE OF GASES

- Provides  $O_2$  for oxidation of food and release of energy.
- Removes  $CO_2$ , a waste product.
- Helps in maintaining pH balance and homeostasis.
- Supports all life processes such as growth, movement, reproduction and repair.
- Maintains proper functioning of cells and organs.

### Quick Revision

- ★ Breathing = Inhalation of  $O_2$  + Exhalation of  $CO_2$
- ★ Exchange of gases = Diffusion between alveoli and blood
- ★  $O_2$  transport = By haemoglobin in RBCs
- ★  $CO_2$  transport = As bicarbonate ions, carbamino compounds and dissolved in plasma
- ★ Regulation = Medulla oblongata (respiratory centre)
- ★ Respiratory pigments = Haemoglobin (humans), Haemocyanin (insects & molluscs)

## 14. SUMMARY DIAGRAM (FLOW OF RESPIRATION IN HUMANS)



$O_2$  diffuses from alveoli → blood |  $CO_2$  diffuses from blood → alveoli

Stay Healthy  
Breathe Clean  
Live Longer

### DOCTOR'S NOTE

Clean air, regular exercise, a balanced diet and no smoking keep your lungs strong and your life long.

Healthy Lungs,  
Happy Life!

### Key Takeaways

- ✓ Understand the process
- ✓ Remember the structures
- ✓ Practice diagrams
- ✓ Revise regularly
- ✓ Stay consistent, stay ahead! ★